RESIDUE FIELD OF A RIG-CLOSED PRIME IS FINITE TYPE

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the ideals of topologically nilpotent elements a in A and b in B for arrows of WAdm^Noeth, and the rig-closed prime notion (a prime q whose closure is contained in a unique maximal ideal m containing b)
- finite type morphisms are stable under composition and restriction of rings (A/a to B/b finite type implies A/a to B/m finite type when b is contained in m)

Lemma 0.1. *Let $\varphi : A \to B$ in $WAdm^{Noeth}$. Denote $\mathfrak{a} \subset A$ and $\mathfrak{b} \subset B$ the ideals of topologically nilpotent elements. Assume $A/\mathfrak{a} \to B/\mathfrak{b}$ is of finite type. Let $\mathfrak{q} \subset B$ be rig-closed. The residue field κ of the local ring $B/\mathfrak{q}$ is a finite type $A/\mathfrak{a}$-algebra.*

Proof. Let $\mathfrak{q} \subset \mathfrak{m} \subset B$ be the unique maximal ideal containing $\mathfrak{q}$. Then $\mathfrak{b} \subset \mathfrak{m}$. Hence $A/\mathfrak{a} \to B/\mathfrak{b} \to B/\mathfrak{m} = \kappa$ is of finite type. □

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